LYU JIAXIN

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SUMMARY

MSc Statistical Data Science student (UCD) seeking biostatistician roles. Experience in regression modelling and model selection, survival analysis, simulation study design and sensitivity analysis in R. Re-analysed a retracted imaging study for my thesis, designed a survival-analysis simulation study, and built tools that turn published evidence into usable output.

EDUCATION

University College Dublin (UCD), MSc Statistical Data Science (GPA 3.54/4.0) *Sep 2025 – Present*

Courses: Survival Models, Mathematical Statistics, MCMC, Machine Learning, Advanced Predictive Analytics, Bioinformatics, Data Programming with R

Harbin University of Commerce, BSc Statistics *Sep 2019 – Jun 2023*

SKILLS

Statistics: linear regression, model selection (AIC), sensitivity analysis and diagnostics (HC3, robust regression), survival analysis (Cox, log-rank, RMST), simulation study design (ADEMP), machine learning (CNN, DNN)

Programming: R (survival, Keras), Python (OpenCV, Streamlit), SQL, SAS

Tools: Git/GitHub, Quarto, Markdown, Excel

Bioinformatics: NCBI protein BLAST, Clustal Omega alignment and phylogenetic trees; alignment and tree-building algorithms (Needleman-Wunsch and Smith-Waterman implemented in R; BLAST, WPGMA, neighbour joining); GenBank, UniProt, PDB

Languages: English (IELTS 6.5 overall), Mandarin (native)

PROJECTS

KETO-CTA Re-Analysis | MSc thesis, sole analyst | R, Python

- Re-analysed the released data of a retracted imaging study (100 participants, 2 CT scans) to test whether baseline plaque and ApoB predict later plaque.
- Compared three regression models by AIC with backward elimination; checked them with log-scale and change-score sensitivity analyses and diagnostics (HC3, robust regression). Baseline plaque held across every model, supporting a qualified version of the original claim.
- Rebuilt participant-level ApoB from a published plot with a Python tool validated against known values (mean error 0.13–0.25 mm^3, under 0.2% of the axis); ApoB added no clear association ($p = 0.82$).

Survival Analysis Simulation | Personal project | R, Quarto

- Designed with the ADEMP framework: 24 scenarios × 1,000 repetitions comparing Cox, log-rank and RMST tests under four effect patterns, two censoring levels and three sample sizes.
- Checked trustworthiness: Type I error 0.043–0.066 against a 0.05 target; no clear RMST bias (under twice the Monte Carlo standard error in all 24 scenarios).
- Found that with crossing curves Cox power rose from 0.33 to 0.97 as sample size grew while RMST stayed near 0.05, so the estimand should be chosen before the test.

Check-up Planner | Self-built tool | JavaScript, bilingual, runs in the browser

- Turned published screening guidelines (USPSTF, ACS, ACG, Chinese guidelines, OSHA and GBZ 188-2025) into a table of 41 rules, each with an evidence level and its sources.
- Graded evidence in five levels: inferences are labelled “Consider”, and nine common tests that guidelines advise against for people without symptoms are listed separately.

Scatter Plot Digitiser | Personal tool | Python, OpenCV, Streamlit

- Recovers data points from a published scatter plot using axis ticks, flags each as automatic or manual, and validates against known values (largest error 1.07 mm^3 on an axis range of about 160 mm^3).

Human Activity Recognition | UCD coursework | R, Keras. Compared a DNN and a CNN on wearable-sensor time series (19 activity classes); the CNN generalised better, with 97.8% test accuracy.

EXPERIENCE

Chengdu VisionVS Technology Co., Ltd., Product Assistant & Delivery Engineer, Chengdu, China *Jul 2023 – Mar 2025*

- Wrote requirements, design specifications and technical documentation with clients for interactive electronic technical manuals (IETM, IETP) and an EHS management system; led an internal project-management system from a survey of four tools to delivery.
- Researched the market and compared three EHS systems (about 196 functions) to define a standard product, and forecast market size and revenue; trained users and worked with developers on iterations.